# Solution: Count of Subset Sum

## Problem Statement

Given a set of positive numbers, find the total number of subsets whose sum is equal to a given number 'S'.

**Example 1**:

```
Input: {1, 1, 2, 3}, S=4
Output: 3
The given set has '3' subsets whose sum is '4': {1, 1, 2}, {1, 3}, {1, 3}
Note that we have two similar sets {1, 3}, because we have two '1' in our input.
```

**Example 2**:

```
Input: {1, 2, 7, 1, 5}, S=9
Output: 3
The given set has '3' subsets whose sum is '9': {2, 7}, {1, 7, 1}, {1, 2, 1, 5}
```

## Basic Solution

This problem follows the **0/1 Knapsack pattern** and is quite similar to `"Subset Sum"`. The only difference in this problem is that we need to count the number of subsets, whereas in the `"Subset Sum"` we only wanted to know if there exists a subset with the given sum.

A basic brute-force solution could be to try all subsets of the given numbers to count the subsets that have a sum equal to 'S'. So our brute-force algorithm will look like:

```
for each number 'i'
  create a new set which includes number 'i' if it does not exceed 'S', and recursively
      process the remaining numbers and sum
  create a new set without number 'i', and recursively process the remaining numbers
return the count of subsets who has a sum equal to 'S'
```

### Code

Here is the code for the brute-force solution:

```java
class Solution {

  public int countSubsets(int[] num, int sum) {
    return this.countSubsetsRecursive(num, sum, 0);
  }

  private int countSubsetsRecursive(int[] num, int sum, int currentIndex) {
    // base checks
    if (sum == 0)
      return 1;

    if(num.length == 0 || currentIndex >= num.length)
      return 0;

    // recursive call after selecting the number at the currentIndex
    // if the number at currentIndex exceeds the sum, we shouldn't process this
    int sum1 = 0;
    if( num[currentIndex] <= sum )
      sum1 = countSubsetsRecursive(num, sum - num[currentIndex], currentIndex + 1);

    // recursive call after excluding the number at the currentIndex
    int sum2 = countSubsetsRecursive(num, sum, currentIndex + 1);

    return sum1 + sum2;
```

Run Code Result | Stdout

You must run your code first

Execute

The time complexity of the above algorithm is exponential $O(2^n)$, where 'n' represents the total number. The space complexity is $O(n)$, this memory is used to store the recursion stack.

## Top-down Dynamic Programming with Memoization

We can use memoization to overcome the overlapping sub-problems. We will be using a two-dimensional array to store the results of solved sub-problems. As mentioned above, we need to store results for every subset and for every possible sum.

### Code

Here is the code:

```java
class Solution {

```

```
    public int countSubsets(int[] num, int sum) {
        Integer[][] dp = new Integer[num.length][sum + 1];
        return this.countSubsetsRecursive(dp, num, sum, 0);
    }


    private int countSubsetsRecursive(Integer[][] dp, int[] num, int sum, int
    currentIndex) {
        // base checks
        if (sum == 0)
        return 1;

        if(num.length == 0 || currentIndex >= num.length)
        return 0;

        // check if we have not already processed a similar problem
        if(dp[currentIndex][sum] == null) {
        // recursive call after choosing the number at the currentIndex
        // if the number at currentIndex exceeds the sum, we shouldn't process this
        int sum1 = 0;
        if( num[currentIndex] <= sum )
```

Run Code Result | Stdout

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## Bottom-up Dynamic Programming

We will try to find if we can make all possible sums with every subset to populate the array `db[TotalNumbers][S+1]`.

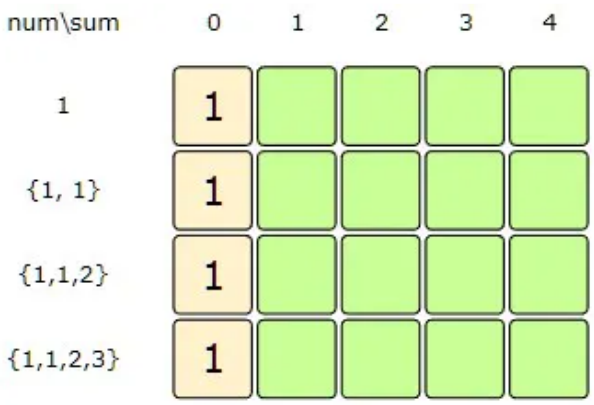
So, at every step we have two options:

1. Exclude the number. Count all the subsets without the given number up to the given sum => `dp[index-1][sum]`
2. Include the number if its value is not more than the 'sum'. In this case, we will count all the subsets to get the remaining sum => `dp[index-1][sum-num[index]]`

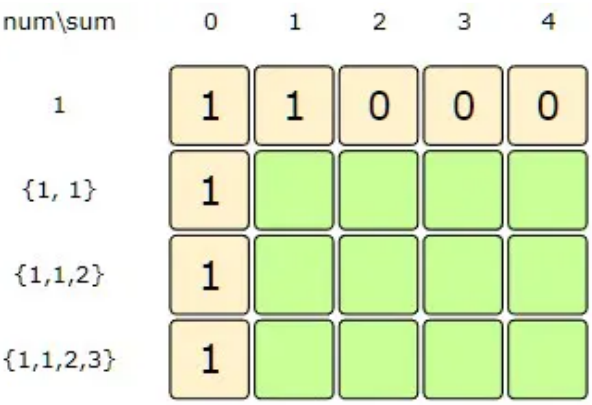
To find the total sets, we will add both of the above two values:

```
    dp[index][sum] = dp[index-1][sum] + dp[index-1][sum-num[index]])
```

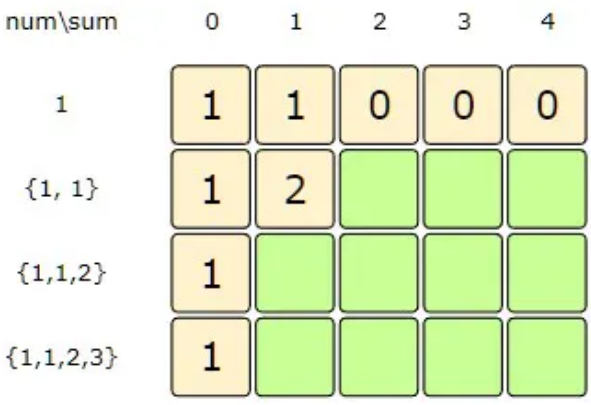
Let's start with our base case of size zero:



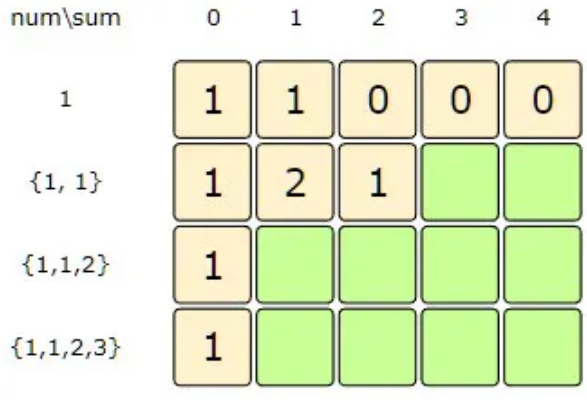

'0' sum can always be found through an empty set




With only one number, we can form a subset only when the required sum is equal to the number




sum: 1, index:1=> (dp[index-1][sum] + dp[index-1][sum - 1])



sum: 2, index:1=> (dp[index-1][sum] + dp[index-1][sum - 1])

| num\sum | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| 1 | 1 | 1 | 0 | 0 | 0 |
| {1, 1} | 1 | 2 | 1 | 0 | 0 |
| {1,1,2} | 1 | 2 | 2 | 2 | 1 |
| {1,1,2,3} | 1 | 2 | 2 | 3 | 3 |

sum: 4, index:3=> (dp[index-1][sum] + dp[index-1][sum - 3])

## Code

Here is the code for our bottom-up dynamic programming approach:

Java

```java
class Solution {

  public int countSubsets(int[] num, int sum) {
    int n = num.length;
    int[][] dp = new int[n][sum + 1];

    // populate the sum=0 columns, as we will always have an empty set for zero sum
    for(int i=0; i < n; i++)
      dp[i][0] = 1;

    // with only one number, we can form a subset only when the required sum is equal to its value
    for(int s=1; s <= sum ; s++) {
      dp[0][s] = (num[0] == s ? 1 : 0);
    }

    // process all subsets for all sums
    for(int i=1; i < num.length; i++) {
      for(int s=1; s <= sum; s++) {
        // exclude the number
        dp[i][s] = dp[i-1][s];
        // include the number, if it does not exceed the sum
        if(s >= num[i])
          dp[i][s] += dp[i-1][s-num[i]];
```

Run Code Result | Stdout

You must run your code first

Execute

The above solution has time and space complexity of $O(N * S)$, where 'N' represents total numbers and 'S' is the desired sum.

## Challenge

Can we further improve our bottom-up DP solution? Can you find an algorithm that has $O(S)$ space complexity?

`Similar to the space optimized solution for "0/1 Knapsack."`

Java

```java
class Solution {
  static int countSubsets(int[] num, int sum) {
    int n = num.length;
    int[] dp = new int[sum + 1];
    dp[0] = 1;

    // with only one number, we can form a subset only when the required sum is equal to its value
    for(int s=1; s <= sum ; s++) {
      dp[s] = (num[0] == s ? 1 : 0);
    }

    // process all subsets for all sums
    for(int i=1; i < num.length; i++) {
      for(int s=sum; s >= 0; s--) {
        if(s >= num[i])
          dp[s] += dp[s-num[i]];
      }
    }

    return dp[sum];
  }

  public static void main(String[] args) {
```

Run Code Result | Stdout

You must run your code first

Execute

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Count Of Subset Sum

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Target Sum

Mark as Completed